Investments
Autumn 2026

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Problem Set #2 Mean-Variance Analysis

Hints for Problem 1: In question 1 of problem set #2 you are asked to calculate the variance and covariance of portfolios. You are asked to calculate these both by using historical data and by using the stock covariances and variances directly. This hint reviews the formulas needed for this second direct approach.

Variance: In question number 1 you are asked to calculate the variance of a portfolio using the information about variances and covariances you calculated using the individual returns in question 1.c. You should have the variance of the return to Target (σ^2_{Target}), variance of the return to Tesla (σ^2_{Tesla}), variance of the return to Alphabet ($\sigma^2_{Alphabet}$), covariance between Target and Tesla ($\sigma_{Target,Tesla}$), covariance between Target and Alphabet ($\sigma_{Target,Alphabet}$), and covariance between Tesla and Alphabet ($\sigma_{Tesla,Alphabet}$). The variance of the portfolio given these is calculated as follows:

$$\begin{aligned}\sigma^2_P =& w^2_{Target}\sigma^2_{Target} + w^2_{Tesla}\sigma^2_{Tesla} + w^2_{Alphabet}\sigma^2_{Alphabet} \\ &+ 2w_{Target}w_{Tesla}\sigma_{Target,Tesla} + 2w_{Target}w_{Alphabet}\sigma_{Target,Alphabet} \\ &+ 2w_{Tesla}w_{Alphabet}\sigma_{Tesla,Alphabet}\end{aligned}$$

where w_{Target} is the weight on Target in the portfolio, w_{Tesla} is the weight on Tesla and $w_{Alphabet}$ is the weight on Alphabet.

Covariance: You are also asked to calculate the covariance between two portfolios using the information on stock variances and covariances. To cut down on the subscripts and superscripts let's consider two portfolios: A and B each invested in three stocks 1, 2 and 3.

When you are using the information on individual stock variances and

covariances, then the covariance between the two portfolios is given by:

$$\begin{aligned}\operatorname{cov}(\tilde{r}_a, \tilde{r}_b) =& w_1^a w_1^b \sigma_{1,1} + w_1^a w_2^b \sigma_{1,2} + w_1^a w_3^b \sigma_{1,3} \\ &+ w_2^a w_1^b \sigma_{2,1} + w_2^a w_2^b \sigma_{2,2} + w_2^a w_3^b \sigma_{2,3} \\ &+ w_3^a w_1^b \sigma_{3,1} + w_3^a w_2^b \sigma_{3,2} + w_3^a w_3^b \sigma_{3,3}\end{aligned}$$